

Clickbait Detection

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1 Problem statement and motivation

With the rapid growth of digital media, users are exposed to an overwhelming amount of online content competing for their attention. Among the various strategies used to maximize engagement, clickbait headlines have become increasingly prevalent. These headlines often rely on curiosity, exaggeration, or ambiguity to attract clicks, but by doing so, they can mislead users and reduce the overall quality of online information. This creates a need for automated systems that can distinguish between clickbait and genuine content.

From a practical perspective, accurate clickbait detection can improve the user experience by filtering misleading content and enhancing the relevance of recommendations. However, detecting clickbait is inherently challenging, as it is not defined solely by specific keywords but also involves subtle linguistic patterns such as sensational phrasing, incomplete information and emotional triggers.

To address this problem, this study develops and evaluates an NLP-based classification pipeline using supervised learning techniques. Specifically, it compares different text representation methods (TF-IDF and Word2Vec) and modelling approaches, including Support Vector Machines and Long Short-Term Memory (LSTM) networks to examine how these choices influence classification performance.

This initiative shapes our research question: How do different text representation techniques and machine learning models influence the effectiveness of clickbait detection? This investigation provides a foundation for evaluating different approaches and understanding their relative strengths.

2 Research hypothesis

Based on the nature of the task, the following hypothesis is proposed: Sequence-based models such as LSTM, combined with appropriate text represen-

tations, will outperform traditional machine learning approaches, as their ability to capture contextual and sequential patterns makes them more effective in identifying the linguistic characteristics of clickbait. This hypothesis is motivated by the observation that clickbait often relies on subtle phrasing, incomplete information, and contextual cues that cannot be fully captured through simple frequency-based or averaged representations. Although traditional approaches such as TF-IDF with SVM are expected to perform well due to their ability to capture discriminative keywords, they may be limited in modeling the structure and flow of text.

3 Related work and background

Early work on clickbait detection has largely framed the problem as a supervised text classification task, where the goal is to distinguish attention-driven headlines from legitimate news content using linguistic and statistical features. (Chakraborty et al., 2016) showed that clickbait headlines exhibit distinctive structural and stylistic patterns, such as exaggerated phrasing and incomplete information, which can be captured through engineered features. Building on this, (Potthast et al., 2016) demonstrated that traditional machine learning models, particularly Support Vector Machines combined with TF-IDF representations, can achieve strong performance by exploiting these lexical cues. These findings suggest that clickbait often follows identifiable patterns at the word and phrase level, making it suitable for feature-based classification approaches.

With the shift towards representation learning in NLP, distributed word embeddings such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) have been widely adopted to capture semantic relationships between words. Unlike sparse representations, these embeddings allow models to generalise across similar words and contexts. However, when used with traditional

classifiers, these embeddings are typically aggregated into fixed-length representations, resulting in the loss of word order and sentence structure. This limitation is particularly important for clickbait detection, where meaning often depends on how phrases are constructed rather than just which words are present.

More recent approaches have focused on neural architectures that can directly model sequential and contextual information in text. Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) are designed to capture dependencies between sequences and have shown strong performance in a range of NLP tasks. In the context of clickbait detection, modern models further incorporate attention mechanisms and stylistic modelling to capture both local and global features of headlines. For example, (sty, 2020) propose a style-aware co-attention framework that jointly models contextual and stylistic information, demonstrating improved performance over traditional approaches. Similarly, broader work on neural text classification (Kim, 2014). (Zhang and Wallace, 2018) highlights the ability of deep learning models to learn richer representations without relying on handcrafted features.

In addition to complex neural models, efficient approaches such as FastText (Joulin et al., 2017) have been proposed to balance performance and computational cost by combining word embeddings with simple architectures. Related research in detecting misinformation and fake news detection (Shu et al., 2017) further emphasises the importance of analysing linguistic patterns in misleading content, reinforcing the relevance of clickbait detection within a broader context.

Despite these advances, existing work often evaluates models in isolation or focuses on specific architectures, making it difficult to directly compare the impact of different representation techniques and modelling choices. In particular, there is limited work that systematically contrasts traditional feature-based methods with embedding-based approaches and sequence models under a unified experimental framework. This study addresses this gap by comparing TF-IDF and Word2Vec representations with both traditional (SVM) and neural (LSTM) models providing a clearer understanding of their relative strengths and limitations for clickbait detection.

3.1 Accomplishments

- Task 1: Pre-process the dataset to a suitable format - Completed.
- Task 2: Tokenise and pad the text data for neural network input - Completed.
- Task 3: Convert text data into TF-IDF representations - Completed.
- Task 4: Build and train an SVM model using TF-IDF features and evaluate its performance - Completed.
- Task 5: Generate Word2Vec embeddings for the dataset - Completed.
- Task 6: Build and train an SVM model using Word2Vec representations and evaluate its performance - Completed.
- Task 7: Design and implement an LSTM model for sequence-based text classification - Completed.
- Task 8: Train and evaluate the LSTM model using different optimisation strategies (Adam and AdamW) - Completed.
- Task 9: Compare the performance of the TF-IDF + SVM, Word2Vec + SVM, and LSTM models - Completed.
- Task 10: Optimise model performance through hyperparameter tuning and early stopping - Completed.
- Task 11: Save trained models and implement a separate test evaluation pipeline for reproducibility - Completed.
- Task 12: Perform error analysis using confusion matrices and misclassified examples to understand model behaviour - Completed.

4 Approach and Methodology

Our approach to clickbait detection integrates both traditional machine learning methods and neural network-based techniques to construct a comparative NLP pipeline. The task is framed as a binary text classification problem, where headlines are classified as clickbait or non-clickbait. To address this, multiple feature representations and model architectures are explored, enabling a systematic

comparison of their effectiveness in capturing the linguistic characteristics of clickbait.

As a baseline, an TF-IDF representation is combined with a Support Vector Machine (SVM) classifier. This approach captures key lexical features and performs well in text classification tasks. However, it cannot capture word order or contextual relationships. Since clickbait often relies on phrasing and sequential patterns, this limitation can lead to misclassification of context-dependent headlines.

To address this, Word2Vec embeddings are explored as an alternative representation. Word2Vec captures semantic relationships by representing words in a continuous vector space. In this project, embeddings are aggregated to form sentence-level representations and used with an SVM classifier. While this improves semantic understanding, averaging embeddings removes word order leading to loss of contextual information and limiting performance.

To overcome these limitations, a Long Short-Term Memory (LSTM) model is implemented to capture sequential dependencies. Unlike traditional models, LSTMs process text as ordered sequences, enabling them to learn contextual patterns across words. This makes them suitable for clickbait detection, where meaning depends on the structure. The LSTM model consists of an embedding layer, a recurrent layer, and a dense output layer. Additionally, two optimisation strategies are compared: Adam and AdamW. Adam provides adaptive learning, while AdamW incorporates weight decay for improved generalisation. Early stopping and hyperparameter tuning are applied to improve performance and prevent overfitting.

All models were implemented within a unified codebase. TF-IDF and Word2Vec models were built using scikit-learn pipelines, while the LSTM model was implemented using Keras. Trained models and preprocessing components, including the tokenizer, were saved and reused during evaluation to ensure consistency and reproducibility.

During implementation, several challenges were encountered. Ensuring consistency between training and testing pipelines was critical, particularly for the LSTM model, where mismatches in tokenisation or padding could affect predictions. This was resolved by reusing the same tokenizer and preprocessing steps. Additionally, tuning the LSTM required careful adjustment of hyperparameters such as learning rate and dropout to balance performance

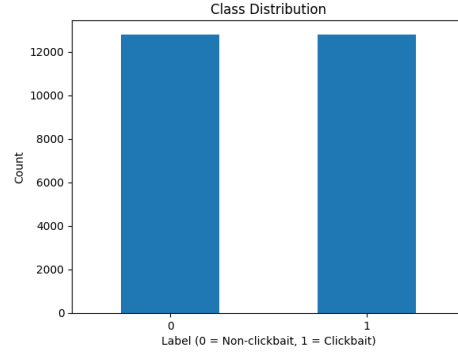


Figure 1: Class distribution of the dataset (0 = Non-clickbait, 1 = Clickbait)

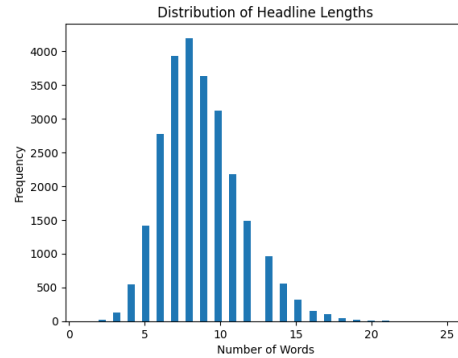


Figure 2: Overall distribution of headline lengths

and overfitting. Despite these challenges, a fully functional pipeline was successfully implemented and evaluated

5 Dataset

The study uses a labelled clickbait dataset consisting of news headlines, where each instance is classified as clickbait (1) or non-clickbait (0). The dataset is balanced, containing 12,801 non-clickbait and 12,799 clickbait instances. The data is split into training and test sets, with no missing values observed. Each instance consists of a short headline and its corresponding label. Clickbait headlines typically use exaggerated or curiosity-driven phrases such as "You won't believe what happened next", while non-clickbait headlines are more factual, for example "Government announces new healthcare policy". These examples highlight that the distinction between classes often depends on subtle linguistic patterns rather than explicit keywords. Figure 1 shows the class distribution, which confirms that both classes are equally represented. Figure 2 illustrates the distribution of the lengths of the headlines, where most headlines contain between 5 and 12 words, on average around

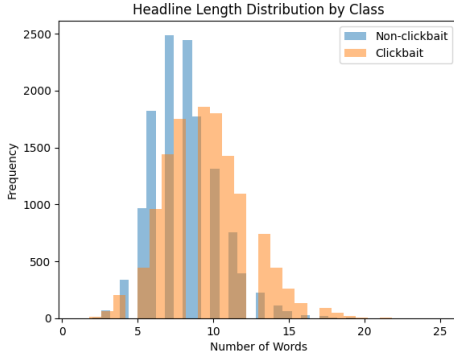


Figure 3: Distribution of headline lengths for clickbait and non-clickbait classes

7-10 words. This indicates that the data set consists primarily of short text sequences. Figure 3 compares the distribution of headline lengths in both classes. The distributions are largely similar, suggesting that headline length alone is not a strong distinguishing feature. As a result, models must rely on linguistic and contextual cues rather than simple statistical properties.

Despite the clear differences in writing style, the task remains challenging. Some non-clickbait headlines contain strong or engaging language, while some clickbait headlines appear relatively neutral. Additionally, the short length of the headlines limits contextual information making accurate classification more difficult.

5.1 Dataset Preprocessing

To prepare the dataset for model training, a series of preprocessing steps were applied to standardise the text while preserving meaningful linguistic patterns. The text data was first converted to lowercase to ensure consistency and reduce vocabulary size. No aggressive preprocessing techniques such as stop word removal or lemmatisation were applied, as clickbait detection often relies on specific phrasing and functional words (e.g., "you", "won't", "this"), which could be lost through such transformations.

For traditional machine learning models, TF-IDF vectorisation was applied directly to the raw text after basic cleaning, allowing the model to capture important lexical features. For the Word2Vec approach, tokenisation was performed to generate word embeddings, which were then aggregated to form sentence-level representations.

For the LSTM model, the text was tokenised using a Keras tokenizer and converted into integer sequences. These sequences were padded to a fixed

length to ensure uniform input size for the model. This step is essential for sequence-based models as it enables batch processing while preserving word order.

One of the main challenges in preprocessing was balancing noise reduction with information preservation. While additional preprocessing steps such as stop word removal could simplify the data, they risk removing important contextual cues that are critical for distinguishing clickbait from non-clickbait. Therefore, a minimal preprocessing approach was adopted to retain as much semantic and structural information as possible.

6 Baselines

To evaluate the effectiveness of the proposed approaches, a baseline model was established using TF-IDF representation combined with a Support Vector Machine (SVM) classifier. This baseline was chosen because TF-IDF is a widely used and effective feature extraction method for text classification tasks, while SVM is known for its strong performance on high-dimensional sparse data.

TF-IDF + SVM baseline provides a reliable reference point for comparison with more advanced approaches such as Word2Vec and LSTM. It captures important lexical features, but does not account for word order or contextual relationships, making it a suitable benchmark for assessing the benefits of more complex models.

Despite its simplicity, the baseline model achieved strong performance, demonstrating that clickbait detection can be partially addressed using surface-level textual features. However, its limitations in handling sequential dependencies motivate the use of neural models such as LSTM, which are better suited to capture contextual patterns in text.

7 Results, error analysis

7.1 Model Performance and Comparison

Table 1 presents the performance of all models across the evaluation metrics.

Model	Acc	Prec	Rec	F1
TF-IDF + SVM	0.9503	0.9613	0.9384	0.9497
Word2Vec + SVM	0.9322	0.9402	0.9231	0.9316
LSTM (Adam)	0.9692	0.9796	0.9584	0.9689

Table 1: Performance comparison of all models

The TF-IDF + SVM model serves as a strong baseline, achieving an accuracy of 0.9503. This in-

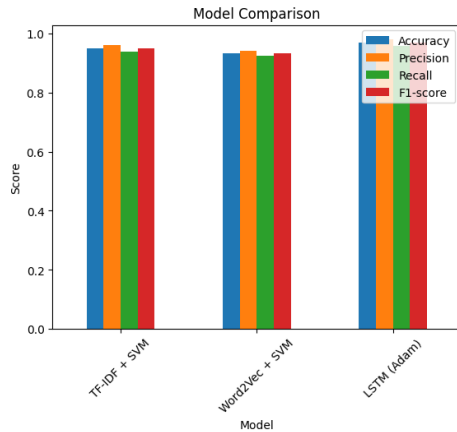


Figure 4: Model Comparison Graph

icates that clickbait detection can be effectively approached using surface-level lexical features. This behaviour aligns with the nature of clickbait headlines, which often contain distinctive words such as "you", "this", or curiosity-inducing phrases. TF-IDF effectively captures these patterns by assigning higher importance to discriminative terms. However, this strength is also a limitation, as the model treats text as a bag of words and ignores word order and context.

The Word2Vec + SVM model achieves lower performance compared to TF-IDF. While Word2Vec embeddings capture semantic similarity between words, the averaging process used to obtain sentence representations removes word order information. This is particularly problematic for clickbait detection, where meaning often depends on phrasing rather than individual words. For example, "you won't believe this" and "this is believed by you" contain similar words but differ significantly in intent. The reduced performance of Word2Vec therefore supports the hypothesis that preserving structure is crucial for this task.

The LSTM model trained with the Adam optimiser achieves the highest performance, with an accuracy of 0.9692. This improvement can be attributed to its ability to model sequential dependencies in text. Unlike TF-IDF and Word2Vec, the LSTM processes text as an ordered sequence, allowing it to capture patterns such as curiosity gaps, incomplete information, and stylistic phrasing commonly used in clickbait. This demonstrates that clickbait detection is not purely a lexical problem but also a structural one.

A comparison between Adam and AdamW optimisers shows that Adam performs slightly better.

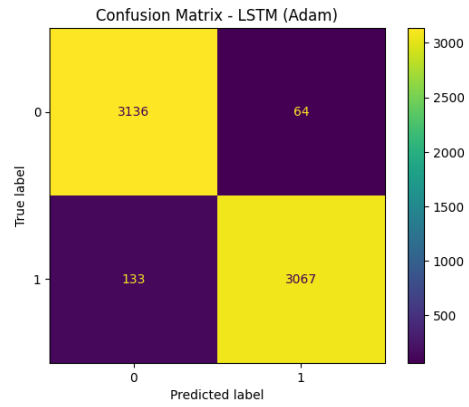


Figure 5: Confusion Matrix of LSTM

The AdamW-based LSTM achieved an accuracy of 0.9683 (F1-score: 0.9680), which is slightly lower than the Adam-based model. Although AdamW introduces weight decay to improve generalisation, the results suggest that this additional regularisation is not necessary for this dataset. The dataset is well-balanced and does not exhibit severe overfitting, which explains why Adam already provides stable and optimal performance.

7.2 Confusion Matrix Analysis

In Figure 5, for the LSTM model, 3136 non-clickbait and 3067 clickbait instances are correctly classified. The number of false positives (64) is relatively low, indicating that the model rarely labels factual headlines as clickbait. However, the number of false negatives (133) is higher, suggesting that the model struggles more in identifying certain types of clickbait.

This pattern indicates that the model is slightly conservative in predicting clickbait. In other words, it requires stronger evidence before classifying a headline as clickbait. This behaviour is beneficial in reducing false alarms, but leads to missed detections when clickbait is subtle.

In comparison, in Figure 6, the TF-IDF + SVM model produces more errors overall, with 121 false positives and 197 false negatives. The higher number of false positives indicates that the model is more sensitive to strong lexical cues, often misclassifying engaging but factual headlines as clickbait. At the same time, the higher number of false negatives suggests that it fails to recognise clickbait that does not contain explicit trigger words.

Across both models, the higher number of false negatives compared to false positives highlights an important characteristic of the dataset: subtle click-

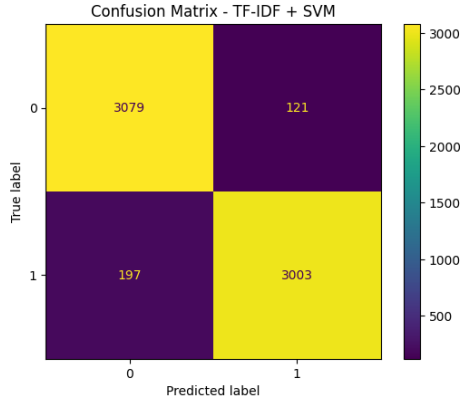


Figure 6: Confusion Matrix of TF-IDF + SVM

bait is inherently more difficult to detect than obvious clickbait or non-clickbait. This observation is consistent with the nature of the task, where the boundaries between classes are not always clearly defined.

7.3 Error Analysis

To gain a deeper understanding of the behavior of the model, a manual error analysis was conducted using misclassified examples from both the LSTM and TF-IDF + SVM models. Rather than relying solely on aggregate metrics, this analysis focuses on understanding why specific errors occur and what they reveal about the limitations of each approach. Table Y presents representative misclassified headlines for both models.

7.3.1 Understanding SVM Errors

The TF-IDF + SVM model demonstrates a clear dependence on lexical features, which directly explains the types of errors observed.

No.	Headline	T	P	Error
1	warner music is singing again	0	1	FP
2	walt disney co reconciles with disney heir	0	1	FP
3	man misdiagnosed as being in coma for years	0	1	FP
4	can you resist the call of the doughnut	1	0	FN
5	are you more elle woods from legally blonde or ...	1	0	FN
6	which cartoons are these villains from	1	0	FN

Table 2: Representative misclassified headlines for TF-IDF + SVM models

In the case of false positives, headlines such as "warner music is singing again" and "man misdiagnosed as being in coma for years", from Table 2, are factual, but contain strong or emotionally engaging words. The model assigns high importance to such terms due to the TF-IDF weighting,

which increases the likelihood of predicting clickbait. However, because the model does not consider how these words are used within the sentence, it does not distinguish between genuine news and stylistically engaging language. This highlights a key limitation: the model captures what words are present but not how they are used.

False negatives further reinforce this limitation. Headlines such as "can you resist the call of the doughnut" and "which cartoons are these villains from" clearly follow common clickbait patterns, particularly through the use of questions and curiosity-driven phrasing. However, these headlines do not necessarily contain strong or distinctive keywords that TF-IDF can prioritise. As a result, the model fails to recognise them as clickbait. This suggests that clickbait detection cannot rely solely on keyword importance and requires modelling of sentence structure.

7.3.2 Understanding LSTM Errors

The LSTM model addresses many of the lexical limitations of TF-IDF by incorporating sequential information, but its errors reveal a different set of challenges.

No.	Headline	T	P	Error
1	forget the cigarette lighter adapter now you can ...	0	1	FP
2	star wars iii premieres at cannes	0	1	FP
3	new romantic comedy film to star drew barrymore ...	0	1	FP
4	bringing up a baby rollsroyce	0	1	FP
5	chrissy teigen is pregnant	1	0	FN
6	new words added to the dictionary today	1	0	FN
7	musicians to add to your new years playlist	1	0	FN

Table 3: Representative LSTM misclassified headlines

False positives in the LSTM model occur when factual headlines resemble clickbait in style. As shown in Table 3, "new romantic comedy film to star drew barrymore" and "star wars iii premieres at cannes" are legitimate news headlines, yet they are predicted as clickbait. These examples contain elements such as named entities and promotional tone, which may resemble patterns learned from clickbait during training. This suggests that the LSTM model is sensitive not only to structure but also to stylistic cues and may overgeneralise patterns associated with attention-grabbing content.

False negatives reveal a more subtle limitation. In Table 3, headlines such as "chrissy teigen is pregnant" and "new words added to the dictionary today" are labelled as clickbait in the dataset but appear neutral and factual. These headlines lack

explicit clickbait signals such as exaggeration or curiosity gaps, making them difficult to distinguish from genuine news. Despite the LSTM's ability to model sequences, it still relies on observable linguistic patterns. When such patterns are weak or absent, the model struggles to make the correct predictions. This indicates that even sequence-based models depend on the presence of identifiable cues and cannot fully resolve the ambiguity.

7.3.3 Comparative Insights Between Models

A direct comparison between the models highlights how their design influences their behaviour. For instance, headlines such as "what to do after you get rejected" and "lies all parents are guilty of telling in public" were misclassified by the SVM model but correctly classified by the LSTM model. These headlines rely on phrasing and implied meaning, rather than explicit keywords, which explains why the LSTM model performs better.

From this comparison, it can be stated that the primary advantage of the LSTM model lies in its ability to capture relationships between words, rather than treating them independently. However, the improvement is not absolute, as both models struggle with ambiguous or borderline cases.

7.3.4 Key Patterns Across Errors

Across both models, several consistent patterns emerge:

- **Ambiguity:** Headlines that are not clearly clickbait or non-clickbait are frequently misclassified.
- **Mild clickbait:** Subtle clickbait without strong exaggeration is often missed (false negatives).
- **Lexical bias:** Models, particularly SVM, are influenced by strong or emotional words, leading to false positives.
- **Stylistic overlap:** Some genuine headlines resemble clickbait in tone, making them difficult to distinguish.
- **Limited context:** The short length of headlines restricts the amount of information available, increasing uncertainty in predictions.

7.4 Discussion

The results confirm that TF-IDF + SVM is a strong and efficient baseline, particularly for capturing surface-level patterns. However, its reliance on

lexical features limits its ability to generalise to more complex linguistic structures.

The Word2Vec approach does not improve performance, mainly due to loss of word order when embeddings are averaged. This highlights that semantic similarity alone is insufficient for clickbait detection, where structure and phrasing are key.

The LSTM model provides clear improvements by capturing sequential dependencies, reducing both false positives and false negatives. However, error analysis reveals that ambiguity remains the primary challenge. When clickbait is subtle or closely resembles genuine headlines, even advanced models struggle to make accurate predictions.

These findings suggest that future improvements may require models that incorporate richer contextual understanding or additional signals beyond text alone. Although LSTM significantly improves performance, it does not fully resolve the challenges associated with ambiguous and borderline cases.

8 Lessons Learned and Conclusion

This study explored multiple approaches for clickbait detection, comparing traditional machine learning models with sequence-based neural networks. As anticipated, the LSTM model achieved the best performance, demonstrating the importance of preserving word order and capturing sequential patterns in text.

A strong baseline was established using TF-IDF + SVM, which performed effectively by capturing lexical patterns. However, its limitations in handling context and structure were evident in the error analysis. In contrast, the Word2Vec + SVM model underperformed, supporting the hypothesis that averaging embeddings leads to loss of structural information critical for this task.

A key insight from this project is that text representation significantly influences model performance. Models that preserve structural and contextual information perform better than those relying solely on word-level features. Additionally, manual error analysis revealed that evaluation metrics alone are insufficient, as models consistently struggled with ambiguous and subtly phrased headlines.

Despite strong overall performance, all models encountered difficulty in handling borderline cases, highlighting the inherent ambiguity in clickbait detection. This suggests that limitations arise not only from model design but also from the nature of

the dataset.

For future work, this study could be extended by evaluating a wider range of models, particularly more advanced architectures capable of capturing richer contextual information. Additionally, using larger and more diverse datasets with a greater variety of headline styles may improve model generalisation and provide a more robust evaluation of performance. Further analysis could also explore how different definitions of clickbait influence classification outcomes, as ambiguity in labelling may contribute to misclassification.

In conclusion, effective clickbait detection depends not only on model choice, but also on selecting representations that align with the structural and linguistic characteristics of the data.

References

2020. Clickbait detection with style-aware title modeling and co-attention. In *CCL*.
- Abhijnan Chakraborty, Bhargavi Paranjape, Sourya Kakarla, and Niloy Ganguly. 2016. Stop clickbait: Detecting and preventing clickbaits in online news media. In *ASONAM*.
- Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural Computation*.
- Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of tricks for efficient text classification. In *EACL*.
- Yoon Kim. 2014. Convolutional neural networks for sentence classification. In *EMNLP*.
- Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.
- Jeffrey Pennington, Richard Socher, and Christopher Manning. 2014. Glove: Global vectors for word representation. In *EMNLP*.
- Martin Potthast, Sebastian Köpsel, Benno Stein, and Matthias Hagen. 2016. Clickbait detection. In *ECIR*.
- Kai Shu, Amy Sliva, Suhan Wang, Jiliang Tang, and Huan Liu. 2017. Fake news detection on social media: A data mining perspective. *SIGKDD Explorations*.
- Ye Zhang and Byron Wallace. 2018. Deep learning for text classification: A survey. *arXiv preprint arXiv:1801.06146*.